Exercises in Theoretical and Empirical Microeconomics and Macroeconomics

Using State-of-the-Art Econometric Methods

Ænnı Snêžàñá Håland-Çałhanőğlü,[1] **Væ-Rie Smārt**,[2] and Äðama Smìß[3]

[1] University of Cologne | [2] University of Bonn | [3] University of Glasgow

June 20, 2026

Outline

1

Introduction

Introduction

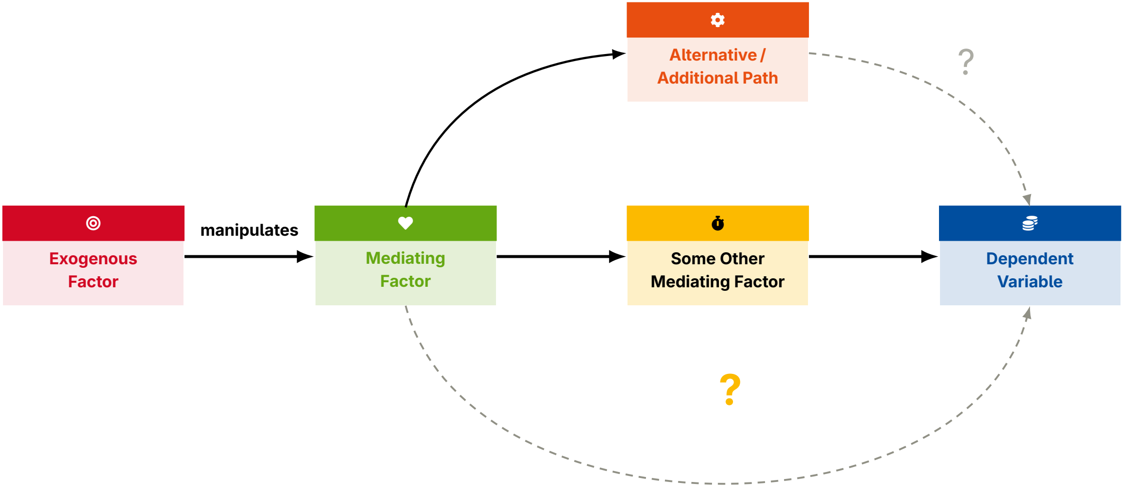
- Theory predicts: Something very specific (Sarfraz and Razzak 2002).
- In practice,
  - we observe this;
  - we also observe that.
- This underlines the importance of analyzing **left-digit bias** (Aristotle 1968).
- *Serif* font *Sans-serif* font *Monospaced* font
- ***Serif* font** ***Sans-serif* font** ***Monospaced* font**

Introduction: Our contribution

1. This is a nested numbered list.
   a) Describe left-digit bias in a different setting: A.
   b) Describe left-digit bias in a different setting: B.
      i. Describe left-digit bias in setting B.1.
      ii. Describe left-digit bias in setting B.2.
      iii. Describe left-digit bias in setting B.3.
      iv. Describe left-digit bias in setting B.4.
2. This is another bullet point.
3. And another bullet point.

Inserting `\par` before the beginning of a lower-level list is necessary for consistent spacing, see https://stackoverflow.com/questions/75483326/latex-beamer-nested-itemize-vertical-spacing-isnt-constant.

A simple schematic



Introduction: Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Introduction: Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Category B

- van Gennep (1909)

Introduction: Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Category B

- van Gennep (1909)

Category C

- Gerhardt (2000), Chiu and Chow (1978)

Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Category B

- Van Gennep (1909)

Related literature

Category A

- Aristotle (1907), Aristotle (1929)

Category B

- Van Gennep (1909)

Category C

- Gerhardt (2000), Chiu and Chow (1978)

No list

Item 1 as plain body text.
Item 2 as plain body text.
With the `beamer` default settings, the text is positioned too close to the top, and the vertical space between paragraphs (items) is too small (namely, absent).
Changing `\parskip` could fix this but has undesirable consequences in other places.
For this reason, one should place *any* text in a `list` environment.

`list` environment

Item 1 in `list` environment.

Item 2 in `list` environment.

Subsequent body text.

`itemize` environment

- Item 1 in `itemize` environment.
- Item 2 in `itemize` environment.

Subsequent body text.

`itemize` environment

- Item 1.
  - Item 1.
  - Item 2.
- Item 2.

Body text.

`enumerate` environment

1. Item 1.
   a) Item 1.
   b) Item 2.
2. Item 2.

Body text.

`enumerate` (1/2)

1. Item 1.
   a) Item 1.
   b) Item 2.

enumerate (2/2)

2. Item 2.
   a) Item 1.
   b) Item 2.

Body text.

Testing `\framebreaks` (1/2)

1. Before frame break.

Testing `\framebreaks` (2/2)

2. After frame break—vertical spacing should be consistent.

Testing `\framebreaks` (1/2)

Before frame break.

Testing `\framebreaks` (2/2)

After frame break—vertical spacing should be consistent.

Testing `\framebreaks` (1/3)

- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.

Testing `\framebreaks` (2/3)

- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.

Testing `\framebreaks` (3/3)

- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.
- Vertical spacing should be consistent.

Questions to Structure Your Presentation (1/3)

You may find the following list of questions helpful as a guide to making sure that your presentation covers all important aspects. The questions equally apply, of course, to posters and manuscripts.

Questions to Structure Your Presentation (2/3)

Your introduction might answer the following questions (not necessarily in this order, but this order will produce a "funnel" structure):

1. What do we already know?
2. What do we not know yet (the "knowledge gap")?
3. Why is it interesting/relevant to close the gap?
4. **What is the research question? [R]**
5. Why is it interesting to answer the research question? ["No one has done it before" ist not sufficient . . .]
6. How do we contribute to closing the knowledge gap: How do we answer the research question? (Which method do we use?)
7. Why is the chosen method (more) suitable (than alternative methods) for answering the research question?

Questions to Structure Your Presentation (3/3)

Your conclusion/discussion might answer the following questions (the following order will produce a "reverse funnel" structure):

8. What was the knowledge gap before the current study? [Similar to the introduction.]
9. **How did we contribute to closing that gap, and what did we find? [A]** Which method/dataset did we use, and what are the results? [Recap and take-home message.]
10. How do our results relate to (confirm/contradict/complement/extend) the results from previous and other contemporaneous studies?
11. What part of the gap is still open? (Phrased a bit more negatively: What are the limitations of our approach?)
12. Next steps/avenues for future research: How could we go about closing the remaining knowledge gap (by removing the limitations of our current approach)?

John Cochrane might do away with question 12. See his "Writing Tips for Ph.D. Students," https://www.fma.org/assets/docs/membercontent/writing_cochrane.pdf.

More Structuring Advice

In her book *The Little Book of Research Writing*, Varanya Chaubey proposes a simpler list of only three items—the **"RAP method"**:

- The **"R"** stands for "research question."
- The **"A"** stands for "answer" (to the research question, of course).
- The **"P"** stands for "positioning statement." (How does our manuscript relate to the literature: Does it corroborate or extend or contradict others' findings?)

Relating the "RAP" approach to the the list of questions above, question 4 is the "R," the answer to question 9 is the "A," and the remaining questions spell out the "P."

For a quick summary of the book, see https://mauve-porcupine-8992.squarespace.com/s/Chaubey_Research_Writing-bxdd.pdf. See also https://www.econscribe.org/about and https://arxiv.org/pdf/2012.07787.

2

Boxes (`block` environments) and columns

`blocks` (1)

This is a standard `block`

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an `alertblock`

Use it for information that you want to stand out even more.

1. Use it for information that you want to stand out even more.
2. Use it for information that you want to stand out even more.

blocks (2)

Theorem (Pythagorean theorem)

In any right triangle, the area of the square whose side is the hypotenuse is equal to the sum of the areas of the squares on the other two sides. Algebraically, $x^2 + y^2 = z^2$.

Definition (Pythagorean triple)

A Pythagorean triple represents the lengths of the sides of a right triangle where all three sides have integer lengths. Such a triple is commonly written (a, b, c).

Examples of Pythagorean triples

Some well-known examples of Pythagorean triples are $(3, 4, 5)$ and $(5, 12, 13)$.

blocks (3)

Lemma

Given two line segments whose lengths are a and b, respectively, there is a real number r such that $b = r\,a$.

Proof.

To prove a lemma by contradiction, try and assume that the statement is false, proceed from there and at some point you will arrive at a contradiction. ■

Using columns and blocks for comparisons

This is a standard `block`

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an `alertblock`

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an `exampleblock`

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

Using columns and blocks for comparisons

This is a standard `block`

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an `alertblock`

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

This is an `exampleblock`

Use it for information that you want to stand out.

- Use it for information that you want to stand out. Use it for information that you want to stand out.
- Use it for information that you want to stand out.

Insurance: What is traded?

We are very used to designing slides linearly from top to bottom.

For instance, we often put (contrasting) juxtapositions in lists—like this:

- **Obvious answer:** a certain payment, the premium, is exchanged for the promise to pay partial or full compensation (cover) for loss resulting from carefully specified events. So premium is exchanged for cover.
- **Less obvious answer:** state-contingent incomes.
  Income is reduced in states of the world in which the specified events do not happen, and increased in states in which the events do happen, as compared to the situation without insurance.

Insurance: What is traded?

It often makes more sense, however, to design (contrasting) juxtapositions by putting the elements side by side—like this:

Obvious answer

A certain payment, the premium, is exchanged for the promise to pay partial or full compensation (cover) for loss resulting from carefully specified events.

So premium is exchanged for cover.

Less obvious answer

State-contingent incomes.

Income is reduced in states of the world in which the specified events do not happen, and increased in states in which the events do happen, as compared to the situation without insurance.

3

Data

Data

- Goal: Identify left-digit bias.
- Bla bla.
- We follow the approach by Goossens et al. (1994):
  - criterion 1,
  - criterion 2,
  - criterion 3.
- Bla bla.
- Bla bla.

▸ Details on the dataset

▸ Dataset in greater detail

Summary statistics

	L–D	D–B	B–L	D–L
Panel A: Description A				
Characteristic 1	46.3 (10.7)	47.6 (10.3)	44.2 (11.5)	43.5 (12.0)
Characteristic 2	97.2 (48.6)	109.9 (90.1)	90.0 (69.2)	78.8 (32.5)
Characteristic 3	7.8 (6.4)	8.7 (6.8)	7.5 (6.2)	7.7 (6.2)
Characteristic 4	16.3 (8.4)	16.9 (8.2)	15.6 (8.0)	14.7 (8.0)
Characteristic 5	11.9 (1.3)	12.3 (1.8)	12.0 (1.3)	11.9 (1.3)
Panel B: Description B				
Characteristic 1	35.0 (9.9)	32.8 (10.0)	35.0 (9.8)	32.8 (10.3)
Characteristic 2	86.5 (28.0)	79.8 (31.3)	91.5 (33.1)	70.6 (25.9)
Characteristic 4	11.4 (7.1)	10.1 (6.6)	11.8 (7.7)	10.2 (6.9)
Characteristic 5	12.0 (1.4)	12.3 (1.9)	12.5 (1.9)	12.0 (1.4)
Number of observations	13,978	1,477	531	1,675

Summary statistics

	L–D	D–B	B–L	D–L
Panel A: Description A				
Characteristic 1	46.3 (10.7)	47.6 (10.3)	44.2 (11.5)	43.5 (12.0)
Characteristic 2	97.2 (48.6)	109.9 (90.1)	90.0 (69.2)	78.8 (32.5)
Characteristic 3	7.8 (6.4)	8.7 (6.8)	7.5 (6.2)	7.7 (6.2)
Characteristic 4	16.3 (8.4)	16.9 (8.2)	15.6 (8.0)	14.7 (8.0)
Characteristic 5	11.9 (1.3)	12.3 (1.8)	12.0 (1.3)	11.9 (1.3)
Panel B: Description B				
Characteristic 1	35.0 (9.9)	32.8 (10.0)	35.0 (9.8)	32.8 (10.3)
Characteristic 2	86.5 (28.0)	79.8 (31.3)	91.5 (33.1)	70.6 (25.9)
Characteristic 4	11.4 (7.1)	10.1 (6.6)	11.8 (7.7)	10.2 (6.9)
Characteristic 5	12.0 (1.4)	12.3 (1.9)	12.5 (1.9)	12.0 (1.4)
Number of observations	13,978	1,477	531	1,675

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table with animated cell highlighting

	Column 2	Column 3	Column 4	Column 5	Column 6
Row 2	(2, 2)	(2, 3)	(2, 4)	(2, 5)	(2, 6)
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3					
Row 4					
Row 5					
Row 6					
Row 7					
Row 8					

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)				
Row 4	(4, 2)				
Row 5	(5, 2)				
Row 6	(6, 2)				
Row 7	(7, 2)				
Row 8	(8, 2)				

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)			
Row 4	(4, 2)	(4, 3)			
Row 5	(5, 2)	(5, 3)			
Row 6	(6, 2)	(6, 3)			
Row 7	(7, 2)	(7, 3)			
Row 8	(8, 2)	(8, 3)			

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)		
Row 4	(4, 2)	(4, 3)	(4, 4)		
Row 5	(5, 2)	(5, 3)	(5, 4)		
Row 6	(6, 2)	(6, 3)	(6, 4)		
Row 7	(7, 2)	(7, 3)	(7, 4)		
Row 8	(8, 2)	(8, 3)	(8, 4)		

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)		
Row 4	(4, 2)	(4, 3)	(4, 4)		
Row 5	(5, 2)	(5, 3)	(5, 4)		
Row 6	(6, 2)	(6, 3)	(6, 4)		
Row 7	(7, 2)	(7, 3)	(7, 4)		
Row 8	(8, 2)	(8, 3)	(8, 4)		

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A tabularray-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	(7, 6)
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	

A `tabularray`-based table, revealed column-/cell-wise

	Group A		Group B		
	Column 2	Column 3	Column 4	Column 5	Column 6
Row 3	(3, 2)	(3, 3)	(3, 4)	(3, 5)	(3, 6)
Row 4	(4, 2)	(4, 3)	(4, 4)	(4, 5)	(4, 6)
Row 5	(5, 2)	(5, 3)	(5, 4)	(5, 5)	(5, 6)
Row 6	(6, 2)	(6, 3)	(6, 4)	(6, 5)	(6, 6)
Row 7	(7, 2)	(7, 3)	(7, 4)	(7, 5)	(7, 6)
Row 8	(8, 2)	(8, 3)	(8, 4)	(8, 5)	(8, 6)

4

Testing various elements

A frame title that is really, really, really, really, really, really, really, really long—works … but strongly discouraged

A frame subtitle—it is also really, really, really, really, really, really, really long. And it works, too. It should still be avoided.

- The coverage chosen is always at a point of tangency between an indifference curve and a budget line, for $q^* > 0$. Thus the cases of **full**, **partial** and **more than full cover** correspond to the cases in which the budget constraint defined by p is respectively **coincident** with, **flatter** than, or **steeper** than the expected value line.
- **Note:** If the budget line is so flat that it does not intersect the indifference curve passing through the initial endowment point, then we have $q^* = 0$.
- **Note:** If only full or zero cover are available, the buyer takes full cover if and only if the resulting point on the certainty line is above the *certainty equivalent* of the initial endowment point at $\tilde{y}$.

Simplest possible model: 2 states (1, 2)

Basic setup for both models | Basic setup for both models | Basic setup for both models

- Income in the no-loss state: $y_1 = y$.
- Income in the loss state: $y_2 = y - L$.
- Probability of loss L is π.
- Expected value of income without insurance:

$$\overline{y} = (1 - \pi)\, y + \pi\, (y - L) = y - \pi L$$

- Expected value of income loss: πL
- Expected utility in the absence of insurance:

$$EU = (1 - \pi)\, u(y) + \pi\, u(y - L)$$

- In the absence of insurance, the individual has an uncertain income endowment.

Methodology: Assumptions on the utility function

- We assume that the preferences over uncertain alternatives are captured by von Neumann–Morgenstern expected utility over income y:

$$EU = \sum_{i=1}^{n} \pi_i \, u(y - L_i) \qquad \boldsymbol{EU = \sum_{i=1}^{n} \pi_i \, u(y - L_i)}$$

or

$$EU = \int_0^{L_{\max}} f(L) \, u(y - L_i) \, \mathrm{d}L \qquad \boldsymbol{EU = \int_0^{L_{\max}} f(L) \, u(y - L_i) \, \mathrm{d}L}$$

for the continuous case.

- u_i is uniquely determined up to positive linear transformation.
- u_i is assumed to be three times differentiable and strictly concave in income y: $u_i'(y) > 0$ and $u_i''(y) < 0$.
- Usually, we assume state independence of utility.

Methodology: Assumptions on the utility function

- We assume that the preferences over uncertain alternatives are captured by von Neumann–Morgenstern expected utility over income y:

$$EU = \sum_{i=1}^{n} \pi_i \, u(y - L_i) \qquad \boldsymbol{EU = \sum_{i=1}^{n} \pi_i \, u(y - L_i)}$$

  or

$$EU = \int_0^{L_{\max}} f(L) \, u(y - L_i) \, \mathrm{d}L \qquad \boldsymbol{EU = \int_0^{L_{\max}} f(L) \, u(y - L_i) \, \mathrm{d}L}$$

  for the continuous case.
- u_i is uniquely determined up to positive linear transformation
- u_i is assumed to be three times differentiable and strictly concave in income y: $u_i'(y) > 0$ and $u_i''(y) < 0$.
- Usually, we assume state independence of utility.

5

Color palettes of various universities from several continents

University of Bonn Color Palette

Colors

Blue
50%

Yellow
50%

Gray
50%

P26 Colors

P26 Yellow
50%

P26 Blue
50%

See https://www.uni-bonn.de/de/universitaet/medien-universitaet/medien-presse-kommunikation/medien-corporate-design/ubo_manual_extern_6_2022.pdf.

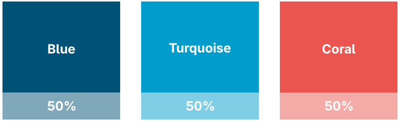
Darker blue (used on uni-bonn.de)

Darker Blue
50%

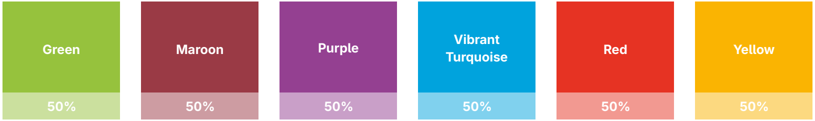
Attention!

University of Cologne Color Palette

Universal colors



Departments’ colors



See https://kommunikation-marketing.uni-koeln.de/e135228/e48647/e49192/e49987/Branding_handbook_eng_ger.pdf.

Attention!

Freie Universität Berlin Color Palette

Primary colors

Black
50%

Green
50%

Secondary color

Blue
50%

Additional colors

Sky Blue
50%

Dusty Sage
50%

Sage
50%

Orange
50%

Burgundy
50%

See https://www.fu-berlin.de/en/sites/corporate-design/guidelines/index.html.

Attention! **Attention!**

TU Dortmund University Color Palette

Primary color

TU Green
50%

Accent color

Orange
50%

Web colors

TU Green muted	Orange muted
50%	50%

See https://itmc.tu-dortmund.de/unsere-services/medien-services/grafik/design/corporate-design/.

Attention!

Ludwig-Maximilians-Universität München Color Palette

Primary colors

LMU Green	LMU Black
50%	50%

Secondary colors

Dark Gray	Mid Gray	Bright Gray	Light Gray
50%	50%	50%	50%

Accent colors (lmu.de)

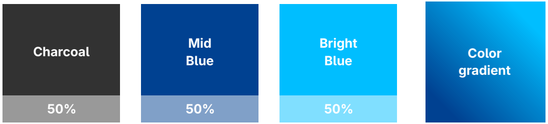
LMU Blue	LMU Cyan	LMU Vilet	LMU Red	LMU Orange
50%	50%	50%	50%	50%

See https://www.lmu.de/de/die-lmu/struktur/zentrale-universitaetsverwaltung/presse-und-kommunikation-puk/lmu-brand-guide/designgrundsaetze/farben/.

Attention!

University of Stuttgart Color Palette

Colors



See https://www.beschaeftigte.uni-stuttgart.de/uni-services/oeffentlichkeitsarbeit/corporate-design/cd-dateien/Uni_Stuttgart_CD-Manual.pdf.

Attention!

Maastricht University Color Palette

Primary color and accent colors

Dark Blue
50%

Orange-Red (accent)
50%

Light Blue (accent)
50%

Bachelor's and master's colors

Orange (bachelor's color)
50%

Red (master's color)
50%

Colors for specific functions

Light Blue alternative
50%

Green (CTA buttons)
50%

Gray (asides)
50%

Light Yellow ("Friendly attention")
50%

See https://www.maastrichtuniversity.nl/support/communications-guide/house-style/colours.

Attention!

Erasmus University Color Palette (1)

Corporate colors

Erasmus Bright Green	Erasmus Green	Erasmus Warm Grey
50%	50%	50%

See https://www.eur.nl/en/about-eur/house-style/brand-elements/colours.

Attention!

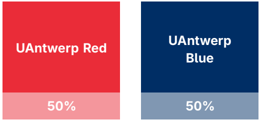
Erasmus University Color Palette (2)

Secondary color palette

Zinc Yellow	Melon Yellow	Traffic Red	Traffic Purple	Luminous Green	Light Blue
50%	50%	50%	50%	50%	50%
Traffic Blue	Pearl Night Blue	Pastel Orange	Grey	Red	
50%	50%	50%	50%	50%	

University of Antwerp Color Palette (1)

Main colors

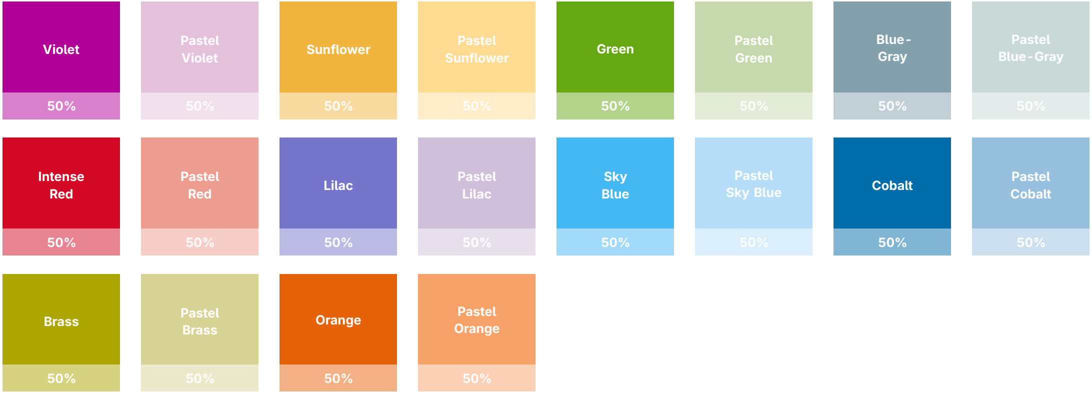


See https://www.uantwerpen.be/en/projects/uantwerp-style-guide/colours/.

Attention!

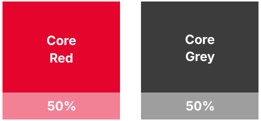
University of Antwerp Color Palette (2)

Faculty colors



University of Leicester Color Palette (1)

Core colors



See https://le.ac.uk/-/media/uol/docs/recruitment/brand-guidelines/uol-quick-guidelines-2020.pdf.

Attention!

University of Leicester Color Palette (2)

Core palette

Orange 50%

Green 50%

Blue 50%

Purple 50%

Web colors

Web Blue 50%

Web Grey 50%

Web Bright Grey 50%

Supporting palette

Bright Yellow 50%

Bright Green 50%

Bright Blue 50%

Bright Purple 50%

Bright Pink 50%

Dark Orange 50%

Dark Green 50%

Dark Blue 50%

University of Oxford Color Palette (1)

Primary color

Oxford Blue

50%

See https://communications.web.ox.ac.uk/communications-resources/visual-identity/identity-guidelines/colours.

Attention!

University of Oxford Color Palette (2)

Secondary colors

Oxford Mauve 50%	Oxford Peach 50%	Oxford Potters Pink 50%	Oxford Dusk 50%	Oxford Lilac 50%	Oxford Sienna 50%
Oxford Red 50%	Oxford Plum 50%	Oxford Coral 50%	Oxford Lavender 50%	Oxford Orange 50%	Oxford Pink 50%

University of Oxford Color Palette (3)

Secondary colors (continued)

Oxford Green 50%

Oxford Ocean Grey 50%

Oxford Yellow Ochre 50%

Oxford Cool Grey 50%

Oxford Sky Blue 50%

Oxford Sage Green 50%

Oxford Viridian 50%

Oxford Royal Blue 50%

Oxford Aqua 50%

Oxford Vivid Green 50%

Oxford Lime Green 50%

Oxford Cerulean Blue 50%

Oxford Lemon Yellow 50%

University of Oxford Color Palette (4)

Neutrals

Oxford Charcoal
50%

Oxford Ash Grey
50%

Oxford Umber
50%

Oxford Stone Grey
50%

Oxford Shell Grey
50%

Oxford Off White
50%

Metallic colors

Gold
50%

Silver
50%

University of Stirling Color Palette (1)

Primary colors

02 Green Heritage Green	01 Green Energy Green	03 Green	01 Green 70%
50%	50%	50%	50%

See https://www.stir.ac.uk/brand-bank/brand-assets/colour-palette/.

Attention!

University of Stirling Color Palette (2)

Secondary colors

03 Teal 50%	02 Teal 50%	01 Teal 50%	01 Teal 70% 50%	03 Blue 50%	02 Blue 50%	01 Blue 50%	01 Blue 70% 50%
03 Purple 50%	02 Purple 50%	01 Purple 50%	01 Purple 70% 50%	03 Pink 50%	02 Pink 50%	01 Pink 50%	01 Pink 70% 50%
03 Orange 50%	02 Orange 50%	01 Orange 50%	01 Orange 70% 50%	03 Yellow 50%	02 Yellow 50%	01 Yellow 50%	01 Yellow 70% 50%

University of Stirling Color Palette (3)

Neutral colors

04 Neutral	03 Neutral	02 Neutral	01 Neutral
50%	50%	50%	50%

Tertiary colors

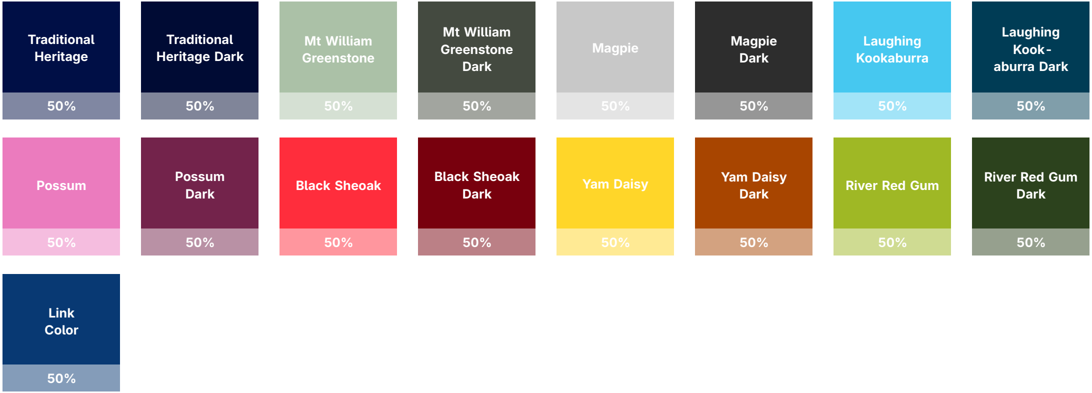
Grey	Bright Grey (website)
50%	50%

University of Stirling Color Palette (4)

Text samples

03 Green	**02** Green/Heritage Green	**01** Green/Energy Green
03 Teal	**02** Teal	**01** Teal
03 Blue	**02** Blue	**01** Blue
03 Purple	**02** Purple	**01** Purple
03 Pink	**02** Pink	**01** Pink
03 Orange	**02** Orange	**01** Orange
03 Yellow	**02** Yellow	**01** Yellow
03 Neutral	**02** Neutral	**01** Neutral
04 Neutral	**00** Grey	

University of Melbourne Color Palette

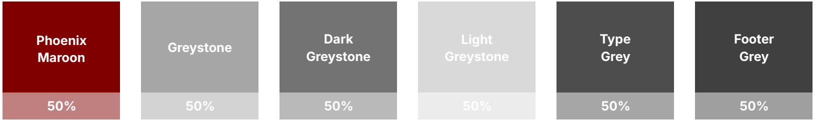


See https://designsystem.web.unimelb.edu.au/style-guide/colour-palette/.

Attention!

University of Chicago Color Palette (1)

Primary digital color palette



See https://uchicago.app.box.com/s/fnrmbycoxhxt4z2ae46ljvo3urmbvlqq.

Attention!

University of Chicago Color Palette (2)

Secondary palette

Light Goldenrod 50%	Light Terracotta 50%	Light Brick 50%	Light Ivy 50%	Light Forest 50%	Light Lake 50%	Light Violet 50%
Goldenrod 50%	Terracotta 50%	Brick 50%	Ivy 50%	Forest 50%	Lake 50%	Violet 50%
Dark Goldenrod 50%	Dark Terracotta 50%	Dark Brick 50%	Dark Ivy 50%	Dark Forest 50%	Dark Lake 50%	Dark Violet 50%

New York University Color Palette (1)

Primary colors

NYU Violet — 50%

Ultra Violet — 50%

Secondary colors

Deep Violet — 50%

Medium Violet 1 — 50%

Medium Violet 2 — 50%

Light Violet 1 — 50%

Light Violet 2 — 50%

See https://www.nyu.edu/employees/resources-and-services/media-and-communications/nyu-brand-guidelines/designing-in-our-style/nyu-colors.html.

Attention!

New York University Color Palette (2)

Neutral colors

Dark Gray	Medium Gray 1	Medium Gray 2	Medium Gray 3	Light Gray
50%	50%	50%	50%	50%

Accent colors

Teal	Magenta	Blue	Yellow
50%	50%	50%	50%

Yale University Color Palette

Blues

Yale Blue — 50%

Blue — 50%

Light Blue — 50%

Grays

Very Dark Gray — 50%

Dark Gray — 50%

Gray — 50%

Light Gray — 50%

Very Light Gray — 50%

Accent colors

Mustard — 50%

Muddy Green — 50%

Burnt Orange — 50%

See https://yaleidentity.yale.edu/guidelines/websites.

Attention!

6

Results

Results

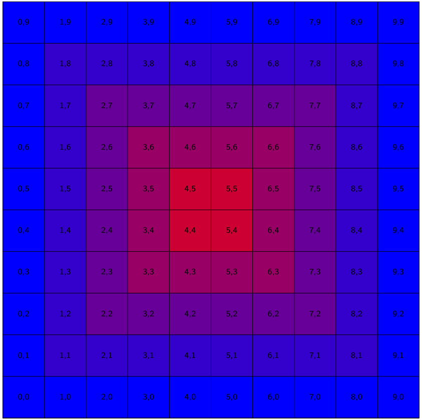


Figure 1. An impressive graph

Results: An `siunitx` example table

Table 1. Example of a regression table

	(1)	(2)	(3)	(4)	(5)
Treatment	−0.390	−0.228	−0.729*	−0.449*	−0.453**
	(+0.352)	(−0.205)	[+0.377]	[−0.245]	{+0.204}
Female	0.948***	0.061	0.188	0.305	0.385*
	(0.354)	(0.233)	(0.372)	(0.226)	(0.222)
Female × Treatment	0.169	0.251	0.892*	0.454	0.439
	(0.514)	(0.325)	(0.533)	(0.341)	(0.307)
Final high school grade	−0.101	0.013	0.076	0.117	0.039
	(0.198)	(0.144)	(0.224)	(0.146)	(0.133)
Trait self-control	−0.016	0.002	−0.016	0.000	−0.007
	(0.016)	(0.010)	(0.015)	(0.010)	(0.009)
Constant	2.357***	1.512***	−0.322	2.158***	1.437***
	(0.239)	(0.144)	(0.265)	(0.161)	(0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment × (1 + Female)	−0.221	0.023	0.163	0.004	−0.014
p_F[Treatment × (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Dependent variable: $m_{\sim}$. Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

Results: An `siunitx` example table

Table 2. Example of a regression table

	(1)	(2)	(3)	(4)	(5)
Treatment	−0.390	−0.228	−0.729*	−0.449*	−0.453**
	(+0.352)	(−0.205)	[+0.377]	[−0.245]	{+0.204}
Female	0.948***	0.061	0.188	0.305	0.385*
	(0.354)	(0.233)	(0.372)	(0.226)	(0.222)
Female × Treatment	0.169	0.251	0.892*	0.454	0.439
	(0.514)	(0.325)	(0.533)	(0.341)	(0.307)
Final high school grade	−0.101	0.013	0.076	0.117	0.039
	(0.198)	(0.144)	(0.224)	(0.146)	(0.133)
Trait self-control	−0.016	0.002	−0.016	0.000	−0.007
	(0.016)	(0.010)	(0.015)	(0.010)	(0.009)
Constant	2.357***	1.512***	−0.322	2.158***	1.437***
	(0.239)	(0.144)	(0.265)	(0.161)	(0.152)
Observations	303	289	295	304	1191
R^2	0.057	0.008	0.039	0.043	0.024
Treatment × (1 + Female)	−0.221	0.023	0.163	0.004	−0.014
p_F[Treatment × (1 + Female) = 0]	0.327	0.008	0.192	0.000	0.003

Dependent variable: $m_{\sim}$. Robust standard errors (cluster-corrected for column 5) in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Missing observations ($N < 308$) due to exclusion of trials in which subjects behaved irrationally (i.e., chose a dominated option). The regressors Final high school grade and Trait self-control are mean-centered.

7

Conclusion

Conclusion

- We have terrific data.
- We have highly convincing findings.
- No-one has done this before
  - Out unique dataset made this possible.
  - Similar attempts: Hostetler et al. (1998), Sigfridsson and Ryde (1998), Markey (2005), Padhye et al. (1999)
- **We are so happy.**

Thank You!

holger.gerhardt@uni-bonn.de
https://www.econ.uni-bonn.de/iame/gerhardt

8

Math “Torture” Test

Math “Torture” Test: Latin and Greek glyphs (1/3)

$\alpha, \beta, \gamma, \delta, \epsilon, , \zeta, \eta, \theta, , \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, , \rho, , \sigma, \varsigma, \tau, \upsilon, \phi, , \chi, \psi, \omega, , A, B, \mathit{\Gamma}, \mathit{\Delta}, E, Z, H, \mathit{\Theta}, I, K, \mathit{\Lambda}, M, N, \mathit{\Xi}, O, \mathit{\Pi}, P, \mathit{\Sigma}, T, Y, \mathit{\Phi}, X, \mathit{\Psi}, \mathit{\Omega}, , - g, \gamma, \mathrm{\gamma}, \mathit{\Gamma}, \mathit{\Gamma}$

$\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, , A, B, \mathit{\Gamma}, \mathit{\Delta}, E, Z, H, \mathit{\Theta}, I, K, \mathit{\Lambda}, M, N, \mathit{\Xi}, O, \mathit{\Pi}, P, \mathit{\Sigma}, T, Y, \mathit{\Phi}, X, \mathit{\Psi}, \mathit{\Omega}, , - g, \gamma, \mathrm{\gamma}, \mathit{\Gamma}, \mathit{\Gamma}$

α, β, γ, δ, ε, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, φ, φ, χ, ψ, ω, , Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, Μ, Ν, Ξ, Ο, Π, Ρ, Σ, Τ, Υ, Φ, Χ, Ψ, Ω, , $- g, \gamma, \mathrm{\gamma}, \mathit{\Gamma}, \mathit{\Gamma}$

$\boldsymbol{\alpha, \beta, \gamma, \delta, \epsilon, , \zeta, \eta, \theta, , \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, , \rho, , \sigma, \varsigma, \tau, \upsilon, \phi, , \chi, \psi, \omega, , A, B, \mathit{\Gamma}, \mathit{\Delta}, E, Z, H, \mathit{\Theta}, I, K, \mathit{\Lambda}, M, N, \mathit{\Xi}, O, \mathit{\Pi}, P, \mathit{\Sigma}, T, Y, \mathit{\Phi}, X, \mathit{\Psi}, \mathit{\Omega}, , - g, \gamma, \gamma, \mathit{\Gamma}, \mathit{\Gamma}}$

$\boldsymbol{\alpha, \beta, \gamma, \delta, \epsilon, \varepsilon, \zeta, \eta, \theta, \vartheta, \iota, \kappa, \lambda, \mu, \nu, \xi, o, \pi, \varpi, \rho, \varrho, \sigma, \varsigma, \tau, \upsilon, \phi, \varphi, \chi, \psi, \omega, , A, B, \mathit{\Gamma}, \mathit{\Delta}, E, Z, H, \mathit{\Theta}, I, K, \mathit{\Lambda}, M, N, \mathit{\Xi}, O, \mathit{\Pi}, P, \mathit{\Sigma}, T, Y, \mathit{\Phi}, X, \mathit{\Psi}, \mathit{\Omega}, , - g, \gamma, \gamma, \mathit{\Gamma}, \mathit{\Gamma}}$

α, β, γ, δ, ε, ε, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, φ, φ, χ, ψ, ω, , Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, Μ, Ν, Ξ, Ο, Π, Ρ, Σ, Τ, Υ, Φ, Χ, Ψ, Ω, , $\boldsymbol{- g, \gamma, \gamma, \mathit{\Gamma}, \mathit{\Gamma}}$

Math “Torture” Test: Latin and Greek glyphs (2/3)

α, β, γ, δ, ε, , ζ, η, θ, , ι, κ, λ, μ, ν, ξ, ο, π, , ρ, , σ, ς, τ, υ, ϕ, , χ, ψ, ω, F, Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, Μ, Ν, Ξ, Ο, Π, Ρ, Σ, Τ, Υ, Φ, Χ, Ψ, Ω, F, — g, γ, γ, Γ, Γ

α, β, γ, δ, ε, ϵ, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω, F, Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, Μ, Ν, Ξ, Ο, Π, Ρ, Σ, Τ, Υ, Φ, Χ, Ψ, Ω, F, — g, γ, γ, Γ, Γ

α, β, γ, δ, ε, ϵ, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω, F, Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, Μ, Ν, Ξ, Ο, Π, Ρ, Σ, Τ, Υ, Φ, Χ, Ψ, Ω, F, — g, *γ*, γ, *Γ*, *Γ*

α, β, γ, δ, ε, , ζ, η, θ, , ι, κ, λ, μ, ν, ξ, ο, π, , ρ, , σ, ς, τ, υ, ϕ, , χ, ψ, ω, F, Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, Μ, Ν, Ξ, Ο, Π, Ρ, Σ, Τ, Υ, Φ, Χ, Ψ, Ω, F, — g, γ, γ, Γ, Γ

α, β, γ, δ, ε, ϵ, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω, F, Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, Μ, Ν, Ξ, Ο, Π, Ρ, Σ, Τ, Υ, Φ, Χ, Ψ, Ω, F, — g, γ, γ, Γ, Γ

α, β, γ, δ, ε, ϵ, ζ, η, θ, ϑ, ι, κ, λ, μ, ν, ξ, ο, π, ϖ, ρ, ϱ, σ, ς, τ, υ, ϕ, φ, χ, ψ, ω, F, Α, Β, Γ, Δ, Ε, Ζ, Η, Θ, Ι, Κ, Λ, Μ, Ν, Ξ, Ο, Π, Ρ, Σ, Τ, Υ, Φ, Χ, Ψ, Ω, F, — g, *γ*, γ, *Γ*, *Γ*

Math “Torture” Test: Latin and Greek glyphs (3/3)

Disambiguation: 0 O O, 1 l I | l I $/$, i j, rn m, ε , θ Θ, φ ψ, ω , $-$ —
Latin vs. Greek: a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, *ß* β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.
Disambiguation: 0 O O, 1 l I | l I $/$, i j, rn m, θ Θ, ϕ ψ, $-$ —
Latin vs. Greek: a α, d δ, e ε, i ι, k κ, n η, o σ, p ρ, *ß* β, u υ, v ν, w ω, x χ, y γ, A Δ Λ, O Θ Ω, T Γ, Y Υ.

Math “Torture” Test: Various expressions and formulas (1/20)

Theorem (Residue Theorem)

Let f be analytic in the region G except for the isolated singularities $a_1, a_2, \ldots, a_m$. If γ is a closed rectifiable curve in G which does not pass through any of the points a_k and if $\gamma \approx 0$ in G, then

$$\frac{1}{2\pi i} \int_\gamma f = \sum_{k=1}^{m} n(\gamma; a_k) \operatorname{Res}(f; a_k).$$

Math “Torture” Test: Various expressions and formulas (2/20)

Theorem (Maximum Modulus)

Let G be a bounded open set in $\mathbb{C}$ and suppose that f is a continuous function on G^- which is analytic in G. Then

$$\max\{\left|f(z)\right| : z \in G^-\} = \max\{\left|f(z)\right| : z \in \partial G\}.$$

Math “Torture” Test: Various expressions and formulas (3/20)

Sample page of mathematical typesetting

First some large operators, both in text: $\iiint_{\mathscr{Q}} f(x,y,z)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ and $\prod_{\gamma\in\Gamma_{\widetilde{C}}}\partial(\widetilde{X}_\gamma)$; and also on display:

$$\begin{aligned}\iiiint\limits_{\mathbf{Q}} f(w,x,y,z)\,\mathrm{d}w\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z &\le \oint_{\partial\mathbf{Q}} f'\left(\max\left\{\frac{\lVert w\rVert}{\lvert w^2+x^2\rvert};\frac{\lVert z\rVert}{\lvert y^2+z^2\rvert};\frac{\lVert w\oplus z\rVert}{\lVert x\oplus y\rVert}\right\}\right)\\ &\precapprox \biguplus_{\mathbb{Q}\Subset\bar{\mathbf{Q}}}\left[f^*\left(\frac{\left\lfloor\int\mathbb{Q}(t)\right\rfloor}{\sqrt{1-t^2}}\right)\right]_{t=\alpha}^{t=}\end{aligned} \tag{1}$$

Math “Torture” Test: Various expressions and formulas (4/20)

For x in the open interval $]-1, 1[$ the infinite sum in Equation (2) is convergent; however, this does not hold throughout the closed interval $[-1, 1]$.

$$(1 - x)^{-k} = 1 + \sum_{j=1}^{\infty} (-1)^j \begin{Bmatrix} k \\ j \end{Bmatrix} x^j \quad \text{for } k \in \mathbb{N};\ k \neq 0. \tag{2}$$

Math “Torture” Test: Various expressions and formulas (5/20)

Most of the following examples are taken from *The TEXbook* (Knuth 1984, see https://ctan.org/pkg/texbook) and were adapted for LATEX from Karl Berry’s torture test for plain TEX math fonts.

Math “Torture” Test: Various expressions and formulas (6/20)

$x+y-z, \quad x+y*z, \quad z*y/z, \quad (x+y)(x-y)=x^2-y^2,$
$x \times y \cdot z = [x\ y\ z], \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,$
$x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y/z, \quad x := y, \quad x \leq y \neq z, \quad x \sim y \simeq z\ x \equiv y \not\equiv z, \quad x \subset y \subseteq z$
$\sin 2\theta = 2 \sin\theta \cos\theta, \quad O(n \log n \log n), \quad \Pr(X > x) = \exp(-x/\mu),$
$\bigl(x \in A(n) \bigm| x \in B(n)\bigr), \quad \bigcup_n X_n \bigm\| \bigcap_n Y_n$

$\boldsymbol{x+y-z, \quad x+y*z, \quad z*y/z, \quad (x+y)(x-y)=x^2-y^2,}$
$\boldsymbol{x \times y \cdot z = [x\ y\ z], \quad x \circ y \bullet z, \quad x \cup y \cap z, \quad x \sqcup y \sqcap z,}$
$\boldsymbol{x \vee y \wedge z, \quad x \pm y \mp z, \quad x = y/z, \quad x := y, \quad x \leq y \neq z, \quad x \sim y \simeq z\ x \equiv y \not\equiv z, \quad x \subset y \subseteq z}$
$\boldsymbol{\sin 2\theta = 2 \sin\theta \cos\theta, \quad O(n \log n \log n), \quad \Pr(X > x) = \exp(-x/\mu),}$
$\boldsymbol{\bigl(x \in A(n) \bigm| x \in B(n)\bigr), \quad \bigcup_n X_n \bigm\| \bigcap_n Y_n}$

Math “Torture” Test: Various expressions and formulas (7/20)

In-text matrices $\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)$ and $\left(\begin{smallmatrix}a&b&c\\1&m&n\end{smallmatrix}\right)$. \$ % ! ? & . , ; :

In-text matrices $\boldsymbol{\left(\begin{smallmatrix}1&1\\0&1\end{smallmatrix}\right)}$ and $\boldsymbol{\left(\begin{smallmatrix}a&b&c\\1&m&n\end{smallmatrix}\right)}$. \$ % ! ? & . , ; :

Math “Torture” Test: Various expressions and formulas (8/20)

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x}\frac{1}{1-x^2} = \frac{a+1}{b} \bigg/ \frac{c+1}{d}.$$

Math “Torture” Test: Various expressions and formulas (9/20)

$$a_0 + \cfrac{1}{a_1 + \cfrac{1}{a_2 + \cfrac{1}{a_3 + \cfrac{1}{a_4}}}}$$

$$\binom{p}{2} x^2 y^{p-2} - \frac{1}{1-x}\frac{1}{1-x^2} = \frac{a+1}{b} \bigg/ \frac{c+1}{d}.$$

Math “Torture” Test: Various expressions and formulas (10/20)

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

Math “Torture” Test: Various expressions and formulas (11/20)

$$\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+\sqrt{1+x}}}}}$$

$$\sqrt[n]{1+\sqrt[k]{1+\sqrt[5]{1+\sqrt[4]{1+\sqrt[3]{1+x}}}}}$$

Math “Torture” Test: Various expressions and formulas (12/20)

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\left|(x + \mathrm{i}y)\right|^2 = 0$$

$$\pi(n) = \sum_{m=2}^{n} \left\lfloor \left(\sum_{k=1}^{m-1} \lfloor (m/k) \rfloor \Big/ \lceil m/k \rceil \right)^{-1} \right\rfloor.$$

$$\int_0^\infty \frac{t - \mathrm{i}b}{t^2 + b^2} e^{\mathrm{i}at}\, \mathrm{d}t = \mathrm{e}^{ab} E_1(ab), \quad a, b > 0.$$

$$\boldsymbol{A} := \begin{bmatrix} x - \lambda & 1 & 0 \\ 0 & x - \lambda & 1 \\ 0 & 0 & x - \lambda \end{bmatrix}.$$

Math “Torture” Test: Various expressions and formulas (13/20)

$$\left(\frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}\right)\left|(x + \mathrm{i}y)\right|^2 = 0$$

$$\pi(n) = \sum_{m=2}^{n} \left\lfloor \left(\sum_{k=1}^{m-1} \lfloor (m/k) \rfloor \Big/ \lceil m/k \rceil \right)^{-1} \right\rfloor.$$

$$\int_0^\infty \frac{t - \mathrm{i}b}{t^2 + b^2} e^{\mathrm{i}at}\,\mathrm{d}t = \mathrm{e}^{ab} E_1(ab), \quad a, b > 0.$$

$$\mathrm{A} := \begin{bmatrix} x - \lambda & 1 & 0 \\ 0 & x - \lambda & 1 \\ 0 & 0 & x - \lambda \end{bmatrix}.$$

Math “Torture” Test: Various expressions and formulas (14/20)

$$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix} \begin{pmatrix} u & x \\ v & y \\ w & z \end{pmatrix}$$

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \ldots & a_{1n} \\ a_{21} & a_{22} & \ldots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \ldots & a_{mn} \end{pmatrix}$$

$$\mathbf{M} = \begin{array}{c} \\ C \\ I \\ C' \end{array} \begin{array}{c} \begin{array}{ccc} C & I & C' \end{array} \\ \begin{pmatrix} 1 & 0 & 0 \\ b & 1-b & 0 \\ 0 & a & 1-a \end{pmatrix} \end{array}$$

Math “Torture” Test: Various expressions and formulas (15/20)

$$\sum_{n=0}^{\infty} a_n z^n \quad \text{converges if} \quad |z| < \left(\limsup_{n\to\infty} \sqrt[n]{|a_n|}\right)^{-1}.$$

$$\frac{f(x+\Delta x) - f(x)}{\Delta x} \to f'(x) \qquad \text{as } \Delta x \to 0.$$

$$\|u_i\| = 1, \qquad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of} \quad \begin{Bmatrix} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{Bmatrix} \quad \text{is} \quad \begin{Bmatrix} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{Bmatrix}.$$

Math “Torture” Test: Various expressions and formulas (16/20)

$$\sum_{n=0}^{\infty} a_n z^n \quad \text{converges if} \quad |z| < \left(\limsup_{n\to\infty} \sqrt[n]{|a_n|} \right)^{-1}.$$

$$\frac{f(x+\Delta x) - f(x)}{\Delta x} \to f'(x) \qquad \text{as } \Delta x \to 0.$$

$$\|u_i\| = 1, \qquad u_i \cdot u_j = 0 \quad \text{if } i \neq j.$$

$$\text{The confluent image of} \quad \begin{Bmatrix} \text{an arc} \\ \text{a circle} \\ \text{a fan} \end{Bmatrix} \quad \text{is} \quad \begin{Bmatrix} \text{an arc} \\ \text{an arc or a circle} \\ \text{a fan or an arc} \end{Bmatrix}.$$

Math “Torture” Test: Various expressions and formulas (17/20)

$$
\begin{aligned}
T(n) \le T(2^{\lceil \lg n \rceil}) &\le c(3^{\lceil \lg n \rceil} - 2^{\lceil \lg n \rceil}) \\
&< 3c \cdot 3^{\lg n} \\
&= 3c\, n^{\lg 3}.
\end{aligned}
$$

$$
\begin{aligned}
(x+y)(x-y) &= x^2 - xy + yx - y^2 \\
&= x^2 - y^2 \\
(x+y)^2 &= x^2 + 2xy + y^2.
\end{aligned}
$$

Math “Torture” Test: Various expressions and formulas (18/20)

$$
\begin{aligned}
\left(\int\limits_{-\infty}^{\infty} \mathrm{e}^{-x^2}\,\mathrm{d}x\right)^2 &= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} \mathrm{e}^{-(x^2+y^2)}\,\mathrm{d}x\,\mathrm{d}y \\
&= \int_0^{2\pi}\int_0^{\infty} \mathrm{e}^{-r^2}\,\mathrm{d}r\,\mathrm{d}\theta \\
&= \int_0^{2\pi}\left(\mathrm{e}^{-\frac{r^2}{2}}\Bigg|_{r=0}^{r=\infty}\right)\mathrm{d}\theta \\
&= \pi.
\end{aligned}
$$

Math “Torture” Test: Various expressions and formulas (19/20)

$$\prod_{k\ge 0} \frac{1}{(1-q^k z)} = \sum_{n\ge 0} z^n \Bigg/ \prod_{1\le k\le n} (1-q^k).$$

$$\sum_{\substack{0<i\le m\\ 0<j\le n}} p(i,j) \ne \sum_{i=1}^{p}\sum_{j=1}^{q}\sum_{k=1}^{r} a_{ij}b_{jk}c_{ki} \ne \sum_{\substack{1\le i\le p\\ 1\le j\le q\\ 1\le k\le r}} a_{ij}b_{jk}c_{ki}$$

Math “Torture” Test: Various expressions and formulas (20/20)

$$\max_{1\le n\le m} \log_2 P_n \quad \text{and} \quad \lim_{x\to 0} \frac{\sin x}{x} = 1$$

Inline math: $\max_{1\le n\le m} \log_2 P_n$ and $\lim_{x\to 0} \frac{\sin x}{x} = 1$

$$p_1(n) = \lim_{m\to\infty} \sum_{\nu=0}^{\infty} \left(1 - \cos^{2m}(\nu!^n \pi/n)\right)$$

Inline math: $p_1(n) = \lim_{m\to\infty} \sum_{\nu=0}^{\infty} \left(1 - \cos^{2m}(\nu!^n \pi/n)\right)$

9

References

References (1/2)

Aristotle. 1907. *De Anima.* Edited by Robert Drew Hicks. Cambridge: Cambridge University Press.

Aristotle. 1929. *Physics.* Translated by P. H. Wicksteed and F. M. Cornford. New York: G. P. Putnam.

Aristotle. 1968. *Poetics.* Edited by D. W. Lucas. Clarendon Aristotle. Oxford: Clarendon Press.

Chiu, Willy W., and We Min Chow. 1978. "A Hybrid Hierarchical Model of a Multiple Virtual Storage (MVS) Operating System." Research report RC-6947. IBM.

Gerhardt, Michael J. 2000. *The Federal Appointments Process: A Constitutional and Historical Analysis.* Durham and London: Duke University Press.

Goossens, Michel, Frank Mittelbach, and Alexander Samarin. 1994. *The LaTeX Companion.* 1st ed. Reading, Mass.: Addison-Wesley.

Hostetler, Michael J., Julia E. Wingate, Chuan-Jian Zhong, Jay E. Harris, Richard W. Vachet, Michael R. Clark, J. David Londono, et al. 1998. "Alkanethiolate gold cluster molecules with core diameters from 1.5 to 5.2 nm: Core and monolayer properties as a function of core size." *Langmuir* 14 (1): 17–30.

Knuth, Donald E. 1984. *The TEXbook.* Vol. A of *Computers & Typesetting.* Reading, Mass.: Addison-Wesley.

Markey, Nicolas. 2005. "Tame the BeaST: The B to X of BibTeX," October 16, 2005. Accessed October 1, 2006. http://mirror.ctan.org/info/bibtex/tamethebeast/ttb_en.pdf.

Padhye, Jitendra, Victor Firoiu, and Don Towsley. 1999. "A Stochastic Model of TCP Reno Congestion Avoidance and Control." Technical report 99-02. Amherst, Mass.: University of Massachusetts.

References (2/2)

Sarfraz, M., and M. F. A. Razzak. 2002. "Technical section: An algorithm for automatic capturing of the font outlines." *Computers and Graphics* 26 (5): 795–804.

Sigfridsson, Emma, and Ulf Ryde. 1998. "Comparison of methods for deriving atomic charges from the electrostatic potential and moments." *Journal of Computational Chemistry* 19 (4): 377–395. https://doi.org/10.1002/(SICI)1096-987X(199803)19:4<377::AID-JCC1>3.0.CO;2-P.

Van Gennep, Arnold. 1909. *Les rites de passage.* Paris: Nourry.

Thank you for your attention—
your feedback is much appreciated!

Thank you for your attention—
your feedback is much appreciated!